From Radiation Fundamentals to Radiotherapy Innovation: A Reflective Account of Undergraduate Research in Radiation Physics and Its Contributions to Bolus Material Science

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ABSTRACT

Background: Radiation therapy remains one of the most effective modalities for cancer treatment. The efficacy of external beam radiotherapy can be substantially enhanced through the use of tissue-equivalent bolus materials, which increase surface dose by mitigating the skin-sparing effect of megavoltage photon and electron beams. However, commercial bolus products are often expensive and inaccessible in resource-limited clinical environments, motivating the investigation of locally available alternatives.

Objective: This reflective article describes a two-year undergraduate research journey in radiation physics at Queensborough Community College (QCC), CUNY. The central focus of the research program was a comparative experimental investigation of Play-Doh and locally sourced materials as low-cost alternatives to commercial Super Flex bolus for beta and gamma radiation attenuation.

Methods: Experiments were conducted using a Geiger-Müller tube, NaI(Tl) scintillation detector, ionization chamber, and semiconductor detectors. Radiation transmission measurements were performed with calibrated sealed radioisotope sources across multiple material samples, including several Play-Doh variants, commercial Super Flex bolus, and honey. Attenuation coefficients and transmission factors were derived from experimental data.

Results: Play-Doh demonstrated radiation attenuation properties comparable to commercial bolus materials under both beta and gamma radiation conditions, consistent with the broader literature on handmade and locally available bolus alternatives. Honey was also evaluated as a novel locally

available candidate. Full experimental results are detailed in the companion manuscript currently under peer review.

Conclusions: Cost-effective, locally available materials such as Play-Doh show promise as bolus alternatives in external beam radiotherapy. The undergraduate research program produced significant scientific and professional outcomes, including a $5,000 CRSP fellowship, the Norman L. Hill Excellence in STEM Research Award, Phi Theta Kappa All-New York Academic Team recognition, Radiation Control Technician (RCT) certification, five conference presentations including the DIA 2025 Global Annual Meeting in Washington, D.C., peer mentorship, and transition into biomedical molecular biology research.

Keywords: bolus material; radiation attenuation; Play-Doh; tissue-equivalent material; external beam radiotherapy; beta radiation; gamma radiation; undergraduate research; CUNY Research Scholars Program

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1. INTRODUCTION

Cancer remains one of the leading causes of mortality worldwide, with an estimated 2.3 million new cases of breast cancer alone diagnosed globally in 2020 [1]. External beam radiotherapy (EBRT) is used in approximately half of all cancer patients at some point during treatment. While the high-energy photon and electron beams used in megavoltage radiotherapy offer excellent deep-tissue penetration, they exhibit a characteristic skin-sparing effect—a dose build-up phenomenon in which the maximum radiation dose is achieved at depth below the skin surface rather than at the surface itself [2,3].

For tumors located at or near the surface—such as post-mastectomy chest wall targets, malignant melanomas, and head-and-neck skin cancers—this skin-sparing effect is therapeutically disadvantageous, as it results in under-dosing of the superficial target region. Tissue-equivalent bolus materials, placed directly on the patient's skin prior to irradiation, shift the 95–100% isodose line toward the skin surface, effectively increasing surface dose and controlling beam penetration depth [3,4,5].

Commercial bolus materials such as Super Flex (SuperFlab) and gel-based polymer products are widely used in clinical radiotherapy. However, these products can be prohibitively expensive,

particularly in community hospitals and low-resource clinical settings [6,7]. This has motivated growing research into locally available, cost-effective alternatives, including beeswax, natural rubber, dental impression material, Play-Doh, paraffin wax, and 3D-printed polymers such as polylactic acid (PLA) and thermoplastic polyurethane (TPU) [7,8,9,10,11,12].

The present article describes a two-year undergraduate research program in radiation physics conducted at the Physics Research Laboratory of Queensborough Community College (QCC), City University of New York (CUNY), under the mentorship of Dr. Rex Taibu. The central research contribution is a comparative experimental study of Play-Doh and locally sourced materials as alternatives to commercial bolus for beta and gamma radiation attenuation. This article also documents the broader educational and professional arc of this research experience—encompassing instrumentation training, professional certification, national honors, international conference participation, peer mentorship, and transition into molecular biology research—to illustrate how community college undergraduate research can produce both scientific and professionally transformative outcomes.

2. BACKGROUND AND LITERATURE CONTEXT

2.1 Bolus Materials in External Beam Radiotherapy

In radiotherapy, bolus materials serve multiple clinical purposes: (i) increasing surface dose in the treatment of superficial malignancies; (ii) compensating for irregular surface contours to achieve dose homogeneity across the target volume; and (iii) modifying beam penetration depth in electron therapy [2,4]. The ICRU Report 24 formally defines a bolus as a tissue-equivalent material that modifies the radiation dose in an irradiated area.

The ideal bolus material should exhibit radiation attenuation properties closely matching those of soft tissue (effective atomic number $Z \approx 7.4$, physical density ≈ 1.0 g/cm^3), be flexible enough to conform to patient surface contours, be non-toxic, durable, and cost-effective [6,13]. Commercial bolus products satisfy many of these criteria but at considerable cost, which may limit availability in community hospitals and clinics in low- and middle-income countries.

The literature documents a broad range of alternative bolus materials. Islam et al. (2020) evaluated the photon attenuation properties of locally sourced beeswax using Monte Carlo MCNP5 simulation and reported good similarity to water and soft tissue across photon energies of 0.015–10

MeV [7]. Nagata et al. (2012) compared Play-Doh, SuperFlab, and wet gauze as electron beam bolus materials and found Play-Doh produced greater attenuation than plastic water, with differences up to 24.7% at 5 MeV [10]. Haryanto et al. (2021) evaluated natural rubber, paraffin wax, red plasticine, and Play-Doh for megavoltage electron and photon therapy and found all materials could serve as soft-tissue substitutes [11]. Separately, dosimetric comparison studies have confirmed that radiation doses resulting from Play-Doh bolus closely approximate those of SuperFlab [15].

Three-dimensional-printed bolus has emerged as a technologically sophisticated alternative. Ono et al. (2023) demonstrated that PLA and TPU 3D-printed bolus materials produced percentage depth-dose (PDD) curves within 3% of RMI457 Solid Water at 6 and 10 MV photon energies [12]. While 3D-printed bolus offers patient-specific customization, it requires specialized equipment not available in most community clinical settings, underscoring the continued relevance of readily available handmade alternatives.

2.2 Radiation Detection Instrumentation

The present study employed radiation detection systems standard in experimental radiation physics. The Geiger-Müller (GM) tube detects ionizing radiation via gas ionization and is widely used for beta and gamma radiation measurements. The NaI(Tl) scintillation detector provides energy-resolved gamma spectroscopy. The ionization chamber is the primary instrument for precise dosimetry, measuring charge produced by ionizing radiation in a defined gas volume. Semiconductor detectors offer high energy resolution for spectroscopic applications [16].

3. RESEARCH JOURNEY AND EXPERIMENTAL PROGRAM

3.1 Phase I: Foundational Radiation Physics (Spring 2024)

The undergraduate research program commenced in January 2024 in the Physics Research Laboratory at QCC under the supervision of Dr. Rex Taibu, Associate Professor of Physics (QCC) and Adjunct Assistant Professor at the New York Institute of Technology (NYIT). The initial phase was dedicated to establishing a comprehensive foundation in radiation detection and measurement.

A systematic program of laboratory experiments was completed using the full suite of available radiation detectors: GM tube, NaI(Tl) scintillation detector, ionization chamber, and semiconductor detector. Early experiments investigated environmental radioactivity by measuring background radiation from natural sources including rocks, soil, seawater, ceramics, and cat litter, as well as radon

gas concentrations. Energy calibration measurements using the NaI(Tl) detector were performed on environmental and food samples, including naturally radioactive foods such as bananas, which contain the potassium-40 (^{40}K) radioisotope.

Controlled experiments using sealed laboratory radioisotope sources included studies of background radiation, resolving time, backscattering, the inverse square law, alpha particle range in air, beta particle absorption, beta decay energy spectra, half-life determination, radiation shielding, and comparative analysis of artificial versus natural radiation sources. These foundational experiments were essential to developing hands-on proficiency with radiation instrumentation and quantitative measurement methodology.

3.2 Phase II: Medical Application Focus and Bolus Research Inception (Spring–Summer 2024)

As experimental competency developed, research focus shifted toward the medical applications of radiation. Participation as a Teaching Assistant in the Introduction to Physics Research (PH-450) course within the College Now program provided early exposure to clinical measurement modalities, including blood pressure monitoring, electrocardiography (ECG), heart rate analysis, and radiation measurement with high school students.

A systematic literature review revealed the central clinical importance of bolus materials in external beam radiotherapy, particularly for superficial tumor treatment. Commercial bolus products were identified as effective but cost-prohibitive in certain clinical environments. Prior studies on the attenuation properties of Play-Doh [10,11,15] suggested its potential as a tissue-equivalent alternative. This led to the formulation of a pilot research project examining Play-Doh as a candidate bolus material under gamma and beta radiation conditions.

This pilot study, titled "Exploring Play-Doh as a Potential Bolus Material in Radiotherapy: Assessing Gamma Radiation and Electron Beam Attenuation Characteristics," was completed and presented at the QCC Spring Student Symposium 2024. Preliminary results supported the hypothesis that Play-Doh could approximate the radiation attenuation behavior of commercial bolus materials.

In Summer 2024, enrollment in the 10-week Radiation Control Technician Training Program at the Universal Training and Testing Academy of America provided intensive professional education in radiation fundamentals, radiation safety protocols, internal and external exposure principles, radiation interactions with matter, dosimetry, and industrial/clinical radiation protection. Successful

completion of the U.S. Department of Energy Core Curriculum Examination with a score of 87% resulted in Radiation Control Technician (RCT) Certification.

During the same summer, service as a High School Research Mentor for the New York City Science Research Mentoring Consortium (NYCSRMC)—a partnership of 28 academic, research, and cultural institutions across New York City that annually connects over 500 scholars with mentored science research experiences [17]—involved supervising three high school students on independent radiation physics projects investigating radiation attenuation, detector efficiency, and backscattering using multiple Play-Doh variants. This mentorship role strengthened leadership, scientific communication, and pedagogical skills.

3.3 Phase III: CUNY Research Scholars Program (Fall 2024–Spring 2025)

In Fall 2024, acceptance into the CUNY Research Scholars Program (CRSP)—the largest mentored undergraduate research experience for associate degree students in the United States, funded by the New York City Mayor's Office and supporting year-long research experiences for 165 associate degree students annually across all seven CUNY community colleges [18]—provided formal funding and institutional support for an advanced research project. The project was titled: "Attenuation of Beta and Gamma Radiation by a Commercial Bolus Material: A Comparative Study with Locally Available Alternatives." CRSP scholars are required to complete 320–400 hours of faculty-mentored research and participate in structured biweekly workshops covering responsible conduct in research, laboratory safety, scientific careers, and presentation skills, culminating in a university-wide summer symposium [18].

The experimental program under CRSP systematically extended the pilot work of Spring 2024. Experiments were conducted comparing multiple Play-Doh samples, commercial Super Flex bolus, and honey as tissue-equivalent materials across multiple radiation detector systems and calibrated measurement geometries. Radiation transmission data were collected for both beta and gamma radiation sources across multiple material thicknesses, from which attenuation coefficients and transmission factors were calculated. The full experimental methodology, quantitative data, and comparative results are presented in the companion peer-reviewed manuscript currently under review [19].

Research findings were presented at two symposia: the QCC Undergraduate Research & Creative Works Day 2024 and the Queens-Wide Undergraduate Research Symposium 2025 at Queens

College, CUNY, under the title "Comparative Analysis of Play-Doh and Super Flex Bolus as Tissue-Equivalent Materials in Radiation Therapy."

During this phase, two major academic honors were received. The Norman L. Hill Excellence in STEM Research Award was conferred for outstanding research and academic achievement in the Physics Department at QCC. Additionally, recognition as a member of the 2025 Phi Theta Kappa All-New York Academic Team was received—awarded for academic excellence, leadership, and community service, with the medallion and certificate presented at the All-New York Academic Awards Ceremony on April 29, 2025 [20].

3.4 Manuscript Preparation and International Dissemination (Summer 2025)

The experimental phase was completed by the end of Spring 2025, and the process of preparing a peer-reviewed manuscript for publication commenced. Manuscript preparation involved comprehensive scientific writing, restructuring to journal standards, figure preparation, and rigorous revision in compliance with author guidelines.

A significant milestone was reached in Summer 2025 when the research was accepted for presentation at the Drug Information Association (DIA) 2025 Global Annual Meeting, held June 15–19, 2025, at the Walter E. Washington Convention Center in Washington, D.C. [21]. DIA is an international conference attended by graduate students, doctoral researchers, medical professionals, regulatory scientists, and industry representatives from the pharmaceutical and biomedical sciences. Participation as one of the few selected undergraduate researchers—alongside PhD candidates from institutions including the University of Oxford—provided a uniquely international platform for dissemination of the bolus material research [21,22].

At the DIA 2025 Global Annual Meeting, participation extended beyond the research poster presentation to include attendance at professional workshops including: "FDA Legislation 2025: How is Congress Preparing to Advance Innovation, Medical Product Safety, and FDA Funding?"; "Leaders of Tomorrow" (focused on building professional networks and identifying career and publication opportunities); and "Novel Framework and Case Study to Support Scientific Rigor: A New Community-Focused Model for Sustainable Clinical Trial Participation." These sessions provided direct exposure to the intersection of regulatory science, pharmaceutical research, and clinical trial methodology [21].

This international participation received media coverage from QCC News [22], and the experience was described as one in which "professionals from the FDA, pharmaceutical companies, biotech industries, publishers, and fellow researchers" were engaged [21]. Dr. Taibu underscored the significance of the invitation, noting that most other selected student presenters were either PhD candidates or graduate-level students [22].

The CRSP research project was subsequently presented at the final CRSP symposium at LaGuardia Community College. The Associate degree was conferred from QCC, and matriculation to The City College of New York (CCNY) for continued undergraduate education in Biology followed.

3.5 Phase IV: Transfer to STEM Student Success Program and Biomedical Research (2025–Present)

In Fall 2025, selection for the CUNY Transfer to STEM Student Success (TS^3) program was achieved. TS^3 is funded by the Alfred P. Sloan Foundation and provides year-long professional development, summer research experiences, and graduate student mentorship for students who have completed associate degrees at CUNY community colleges and transferred to one of four participating four-year institutions: City College, Lehman College, Hunter College, or Queens College [23]. TS^3 students receive a $7,000 stipend for participating in a full year of the program, which culminates in a university-wide summer research symposium [23]. The program supports 32 STEM transfer students annually and is designed to prepare diverse students, including those from groups traditionally underrepresented in STEM, to pursue graduate degrees and STEM careers [23].

During Fall 2025, the CRSP research manuscript was finalized and submitted for peer review. Participation in TS^3 workshops on laboratory safety, science communication, and professional development provided additional preparation for advanced research.

In Spring 2026, a new research direction was initiated by joining the laboratory of Dr. Prabodhika Mallikaratchy, Professor in the Department of Cellular and Biomedical Sciences at the CUNY School of Medicine. Current research focuses on nucleic acid therapeutics, molecular targeting, and innovative cancer treatment strategies. This transition integrates the foundational expertise in radiation physics with molecular biology and biomedical science, establishing a trajectory toward internal radiotherapy, targeted molecular therapy, and translational cancer research—a convergence of disciplines supported by the Biology undergraduate curriculum at City College of New York.

4. RESEARCH CONTRIBUTIONS AND SCIENTIFIC SIGNIFICANCE

4.1 Experimental Framework

The primary experimental study employed a rigorous quantitative framework grounded in established radiation physics measurement methodology. Beta radiation measurements utilized a GM tube with calibrated source-to-detector geometry, and gamma radiation measurements utilized the NaI(Tl) scintillation detector and ionization chamber. Material samples were prepared in standardized geometries, and transmission measurements were conducted at fixed source-to-detector distances with and without intervening bolus materials.

The linear attenuation coefficient (μ) for each material was extracted from the exponential attenuation law $I = I_0 e^{(-\mu x)}$, where I is transmitted intensity, I_0 is incident intensity, and x is material thickness. The mass attenuation coefficient (μ/ρ) was calculated by normalizing μ by the measured physical density (ρ) of each material. These parameters were then compared between Play-Doh variants, honey, and the commercial Super Flex bolus reference material.

4.2 Significance of Play-Doh as a Bolus Candidate

The selection of Play-Doh as the primary candidate was motivated by practical and physical considerations. Play-Doh is commercially available at very low cost, is non-toxic, is malleable and conformable to irregular surface geometries, and requires no specialized fabrication. Prior literature noted its informal clinical use as a bolus material [11], and dosimetric comparison studies confirmed that Play-Doh doses closely approximate those of SuperFlab [15].

However, prior studies also reported that Play-Doh produced greater attenuation than plastic water at lower electron beam energies, with differences as large as 24.7% at 5 MeV [10]. This finding indicates that Play-Doh's attenuation properties may not be perfectly tissue-equivalent across all energy ranges and that machine-specific calibration curves would be required prior to clinical use. The present study's investigation of Play-Doh for beta and gamma radiation extends this characterization to lower-energy radiation regimes relevant to sealed source applications.

4.3 Honey as a Novel Locally Available Alternative

The inclusion of honey as a candidate bolus material represents an exploratory contribution of this research program. Honey is a natural biomaterial with a density of approximately 1.38–1.45 g/cm³ and a complex elemental composition dominated by carbon, hydrogen, and oxygen—broadly similar

elemental constituents to those of soft tissue. The potential for honey or honey-gel composites to serve as bolus material has not been extensively characterized in the published literature, and the present study provides preliminary experimental data on its gamma and beta attenuation properties relative to Super Flex bolus.

5. EDUCATIONAL AND PROFESSIONAL DEVELOPMENT OUTCOMES

The undergraduate research program documented here produced a range of educational and professional outcomes that extend well beyond the scientific findings themselves.

Academic Honors and Recognition: The Norman L. Hill Excellence in STEM Research Award was received for outstanding research and academic achievement in the Physics Department at QCC. Recognition as a member of the 2025 Phi Theta Kappa (PTK) All-New York Academic Team—the internationally acclaimed honor society for two-year college students—was awarded for demonstrated academic excellence, leadership, and community service [20]. PTK induction into the Lambda Sigma Chapter at QCC further recognized sustained academic achievement.

Institutional Funding: The CUNY Research Scholars Program (CRSP) fellowship provided $5,000 in research funding, 320–400 hours of mentored research, and structured professional development over the 2024–2025 academic year. The TS³ program subsequently provided $7,000 in funding for the 2025–2026 academic year, supporting research mentorship and professional development at City College of New York [23].

Professional Certification: The Radiation Control Technician (RCT) Certification, awarded upon passing the U.S. Department of Energy Core Curriculum Examination with a score of 87%, provides a nationally recognized professional credential in radiation safety and protection.

High School Peer Mentorship: Supervision of three high school students on independent radiation physics projects through the NYCSRMC—a consortium of 28 partner programs connecting over 500 scholars annually with mentored science research experiences [17]—developed scientific leadership and broadened physics education at the pre-college level.

Conference Presentations (5 total): Research was presented at five venues: (1) QCC Spring Student Symposium, 2024; (2) QCC Undergraduate Research & Creative Works Day, 2024; (3) Queens-Wide Undergraduate Research Symposium at Queens College, CUNY, 2025; (4) CRSP Final Symposium

at LaGuardia Community College, 2025; and (5) DIA 2025 Global Annual Meeting, Walter E. Washington Convention Center, Washington, D.C., June 15–19, 2025 [21]—an international conference at which most other student presenters were doctoral candidates or graduate students.

Media Coverage: QCC News published a feature article documenting the DIA 2025 presentation and the broader research journey, highlighting the significance of an undergraduate researcher participating alongside doctoral candidates from institutions including the University of Oxford [22].

Transition to Biomedical Research: The radiation physics research program directly enabled transition into molecular biology research at the CUNY School of Medicine, demonstrating how community college undergraduate research can serve as a launching point for interdisciplinary scientific careers at the interface of physics, biology, and medicine.

6. DISCUSSION

The research trajectory described in this article demonstrates that community college undergraduate researchers can produce scientifically meaningful contributions to applied radiation physics when provided with appropriate mentorship, laboratory access, and institutional support. The bolus material investigation addresses a genuine clinical need—the availability of cost-effective, tissue-equivalent materials for external beam radiotherapy—that has direct implications for patient care in resource-limited settings.

The experimental finding that Play-Doh exhibits radiation attenuation properties comparable to commercial bolus materials for beta and gamma radiation is consistent with prior literature [10,11,15] and extends earlier characterization to the specific energy range and radiation types investigated in this study. The exploration of honey as a novel candidate adds preliminary data to the emerging literature on locally available bolus alternatives. Neither investigation has required equipment beyond what is available in a well-equipped undergraduate physics teaching laboratory, underscoring the accessibility of this research paradigm for community college settings.

Several limitations merit acknowledgment. The experimental work was conducted using sealed radioisotope sources rather than clinical linear accelerators (LINACs) or electron beam units, which limits direct extrapolation to high-energy megavoltage treatment conditions. Additionally, measurements of percentage depth dose (PDD) and percentage surface dose (PSD) in phantom geometries—standard characterization metrics in clinical bolus studies [3,11,12]—were beyond the

scope of the current laboratory facility. Future work employing LINAC-based measurements in solid water or water-equivalent phantoms would substantially strengthen the clinical relevance of these findings.

The interdisciplinary arc of this research program—from radiation instrumentation, environmental radioactivity, and medical physics, to regulatory science exposure at DIA 2025, and ultimately to molecular biology and nucleic acid therapeutics—illustrates the potential for physics undergraduate research to serve as a gateway to multiple scientific disciplines. The concurrent pursuit of a Biology degree at City College of New York provides a rigorous cellular and molecular biology foundation that complements the physics and medical physics background, positioning this research trajectory toward future work at the interface of targeted radiotherapy and molecular medicine.

The institutional programs that scaffolded this research journey—CRSP, NYCSRMC, the RCT Training Program, and TS³—each contributed distinct elements: funding and structured research time (CRSP); pre-college scientific leadership experience (NYCSRMC); professional credentialing (RCT Certification); and graduate-student mentorship and senior-college integration (TS³). Together they illustrate how a layered ecosystem of undergraduate support programs can produce exceptional research and professional outcomes, even beginning at the community college level.

7. CONCLUSION

This article has presented a reflective account of a two-year undergraduate research program in radiation physics at Queensborough Community College, CUNY. The central scientific contribution is a comparative experimental investigation of Play-Doh, honey, and commercial Super Flex bolus for beta and gamma radiation attenuation, conducted under the CUNY Research Scholars Program. Preliminary and published findings support the potential of Play-Doh as a cost-effective bolus alternative in external beam radiotherapy, consistent with the existing literature.

Beyond the scientific findings, the program produced substantial educational and professional outcomes: the Norman L. Hill Excellence in STEM Research Award; Phi Theta Kappa All-New York Academic Team recognition; a $5,000 CRSP fellowship supporting 400 hours of mentored research; professional certification as a Radiation Control Technician; supervision of high school researchers through the NYCSRMC; five conference presentations across local, regional, and international venues, culminating in the DIA 2025 Global Annual Meeting in Washington, D.C.; and selection for

the Sloan Foundation-funded TS³ transfer research program at City College of New York. These outcomes collectively demonstrate that undergraduate research, even at the community college level, can achieve both scientific rigor and significant professional impact.

The author is currently pursuing a Biology degree at City College of New York and conducting research in molecular biology at the CUNY School of Medicine. Future directions include LINAC-based characterization of Play-Doh and honey attenuation at clinical megavoltage energies, Monte Carlo simulation of photon and electron interactions in candidate materials, and exploration of the intersection of radiation physics and nucleic acid therapeutics in the context of internal radiotherapy and targeted molecular cancer treatment strategies.

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DECLARATION OF COMPETING INTEREST

The author declares no competing financial or personal interests that could have influenced the research described in this article.

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